

Sensation

CT

Troubleshooting Guide

PDC

Troubleshooting Guide PDC

This document is valid for:
SOMATOM Sensation 10/16/Cardiac
SOMATOM Sensation 40/64/Cardiac 64/Open

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1	General	4
	Safety information.	4
	Tools	6
2	PDC	7
	Overview.	7
	Direct Current Circuit (DCC).	8
	Flowchart “Troubleshooting the DCC”.	9
	Additional hints	9
	Power Distribution System (PDS).	11
	Fuse	11
	Service switch.	12
	Other components	13
	Main breaker functionality test	15
	Additional hints	16
3	Changes to previous version	18
4	Index	19

Safety information

Read the following instructions as well as the most recent version of

- General Safety Notes (TD00-000.860.01.xx.xx)
- Product- specific safety Information (CT02-023.860.01.xx.xx)

before trouble shooting any parts inside the PDC (Power Distribution Cabinet)!

The following instructions and the above mentioned **Safety Notes** contain important safety information as well as information about tools to be used.

WARNING

[hz_serdoc_F13G01U12M03]

Avoid accident and injury or damage of parts.

Risk of accident and injury!

- ⇒ Read and observe the safety information contained in the “General” section of this document and/or “Product-specific safety notes”.

CAUTION

[hz_serdoc_F13G01U03M01]

Not following X-rays protective regulations may lead to radiation exposure for yourself and/or other persons.

Risk of radiation exposure!

Observe radiation protection regulations when X-rays are switched on, e.g. :

- ⇒ Never work inside the gantry room.
- ⇒ Do not leave the system unattended.
- ⇒ Make sure there is no one inside the gantry room.
- ⇒ Make sure that no person can enter the gantry room unnoticed (e.g. lock doors if necessary) .

⚠ WARNING

[hz_serdoc_F13G01U05M03]

Dangerous voltages (up to 560 VDC) are present when system is switched on. Dangerous voltages (up to 560 VDC) may be present even when the system is switched off due to capacity power.

Risk of electric shock!

- ⇒ **Observe power-off instructions and discharge wait times (allow at least five minutes discharge time after the last scan for all involved HV and UDC parts).**
- ⇒ **Secure the system against unintended switch-on (e.g. block breaker against switching on and/or mark breaker against switch on).**
- ⇒ **Ensure via measurements that all voltages are switched off.**
- ⇒ **Connect the stationary and rotating parts of the gantry to a protective conductor prior to working in the gantry.**

⚠ WARNING

[hz_serdoc_F13G01U11M01]

Use of service-enable switch (table) disables collision protection.

Risk of accident and injury!

- ⇒ **Be careful when moving the table.**

Tools

- Standard tool kit

Overview

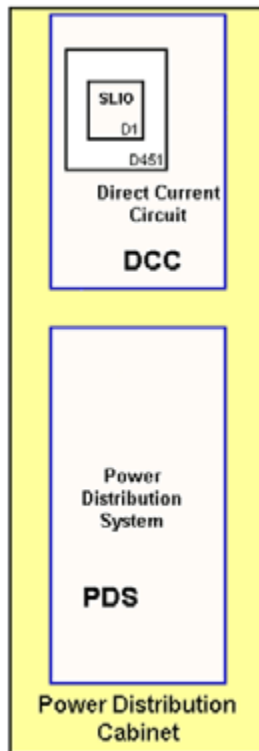


Fig. 1: PDC overview

With regard to function, the PDC can be divided into two separate functions:

- DCC (Direct Current Circuit) located in the upper part of the PDC
- PDS (Power Distribution System) located in the lower part of the PDC

The system may be delivered with the following components:

- **Transformer T100** if the line voltage is not equal to 400 V
- UPS (Uninterrupted Power Supply) - Option

Direct Current Circuit (DCC)

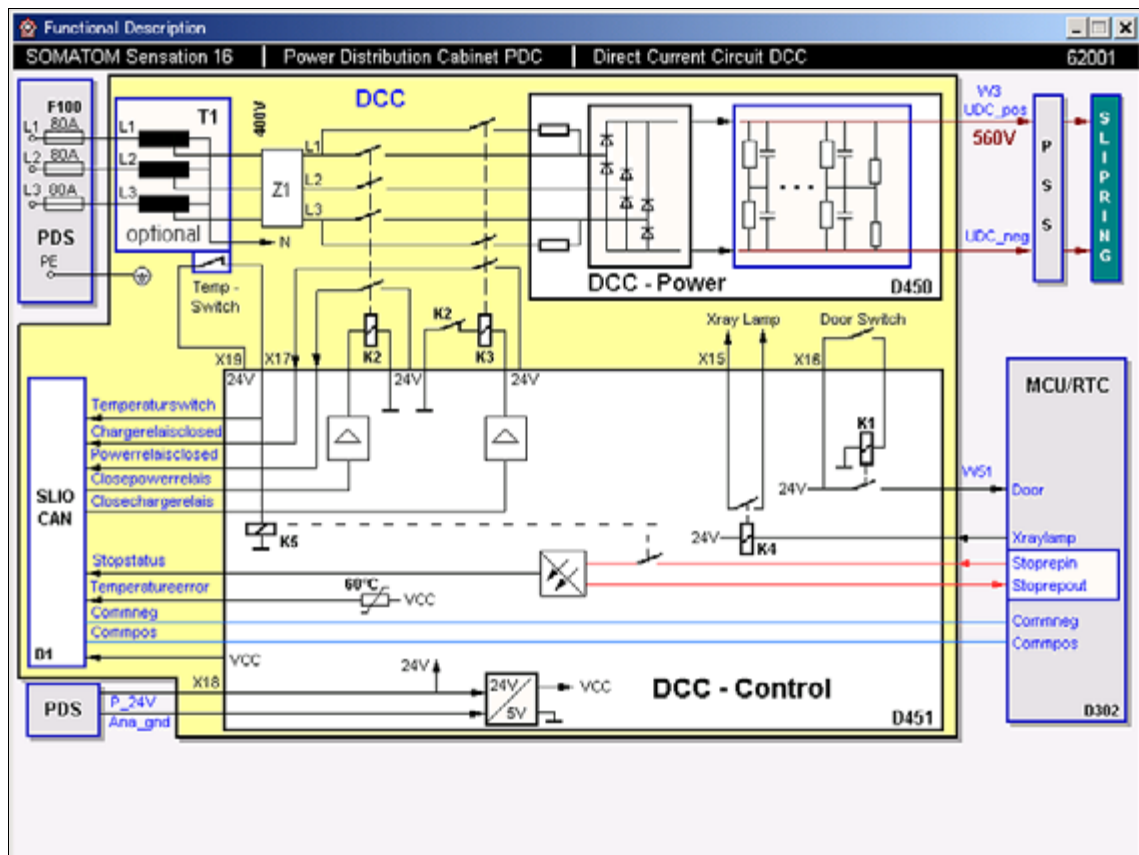


Fig. 2: DCC overview

The DCC supplies the Generator electronic box (GEL) with the required UDC voltage of approx. 560V (UDC_p and UDC_n). These voltage are monitored at the main control board (D400) of the GEL. A missing DCC voltage will generate an error message, e.g CT_XRS_3023 or CT_XRS_3054.

Please use (Fig. 3 / p. 9) to process in the event that

- the system does not reach the "Reset" or "Stand by" state
- the error messages CT_XRS_3023 or CT_XRS_3054 were generated.

Flowchart "Troubleshooting the DCC"

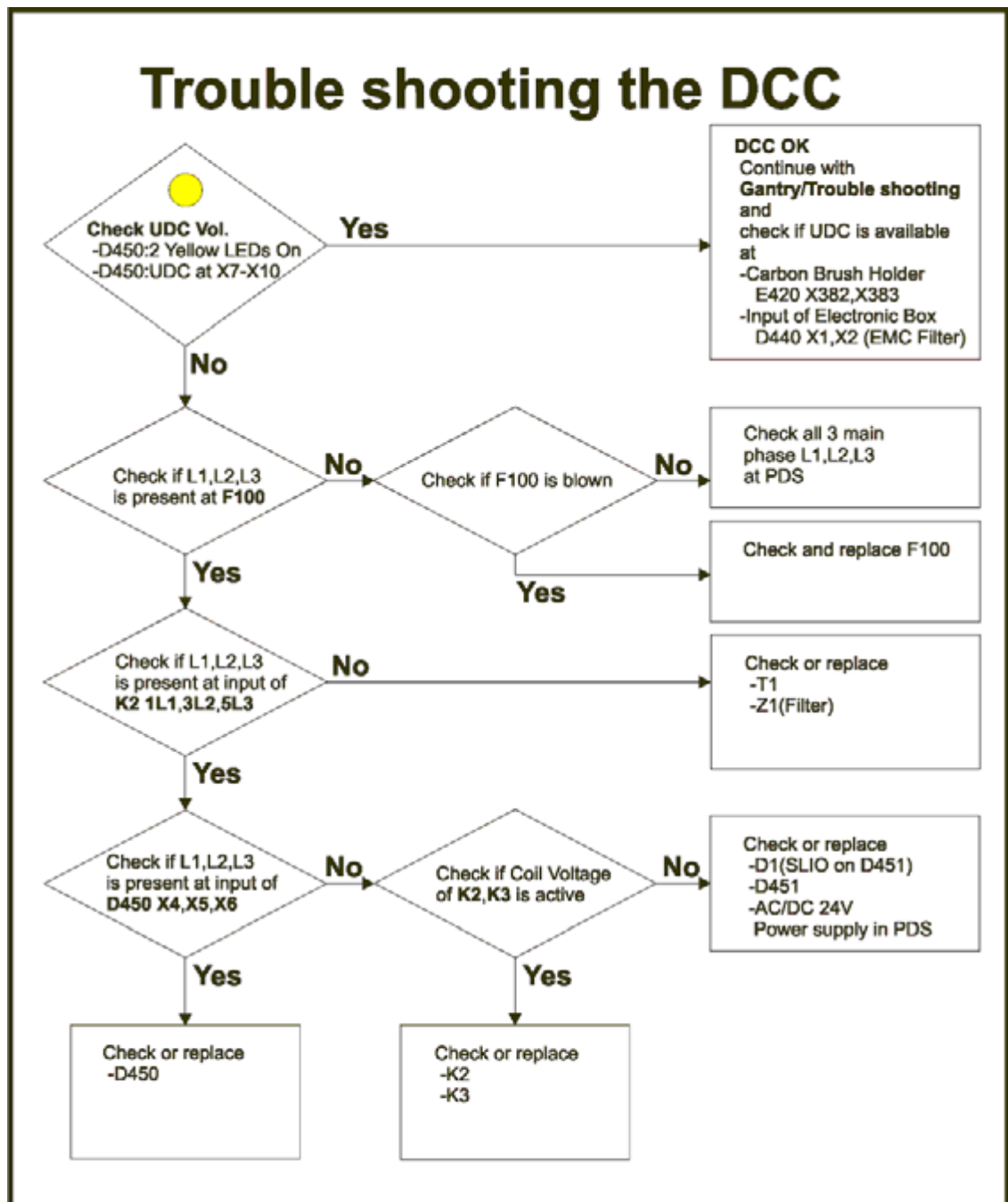


Fig. 3: Flowchart "Troubleshooting the DCC"

Additional hints

- The DCC is controlled by a SLIO(D1) on the DCC Control (D451), error messages are shown as CT_XRS errors.
- If one of the F100 fuses is defective, all three fuses have to be replaced!

- **T100:**

If an additional Transformer T100 is not installed (necessary if line voltage is not equal to 400 V), the temperature switch of T100 has to be shortened on D451 X19.

Power Distribution System (PDS)

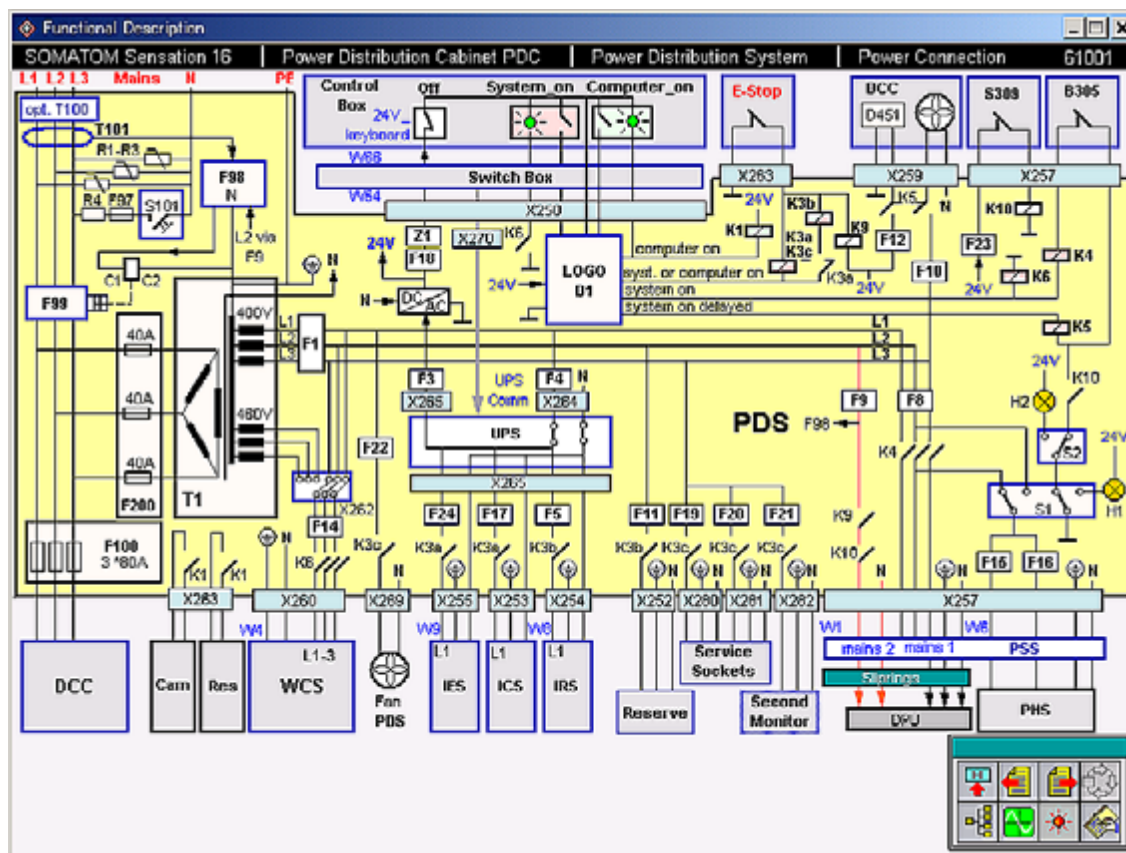


Fig. 4: PDS overview

The PDS distributes the hospital main power to all system components. As a result, a fault in the PDS can generate

- no error message
- various error messages from different system components (controller).

This section gives some additional troubleshooting information, in case the information in the detailed message or functional description is insufficient.

Fuse

The following table shows the fuses required to switch on some system components independently:

Tab. 1 PDC load

Load	Required fuses	Required relays	Con- nection	Comp. ON	Sys. ON	Site On
Control Box 24VDC	F1, F4, F3, F18	none	n.a.	no	no	yes
IRS 230VAC L1	F1, F4, F5	K3b	X254	yes	yes	no

Load	Required fuses	Required relays	Con- nection	Comp. ON	Sys. ON	Site On
ICS 230VAC L1	F1, F4, F17	K3a	X253	yes	yes	no
PDC fan 230VAC L1	F1, F22	K3c	X269	yes	yes	no
2nd Monitor 230VAC L3	F1, F21	K3c	X282	yes	yes	no
Service socket1 230VAC L3	F1, F19	K3c	X280	yes	yes	no
Service socket2 230VAC L3	F1, F20	K3c	X281	yes	yes	no
Reserved 230VAC L2	F1, F11	K3c	X252	yes	yes	no
Gantry 230VAC	F1, F8	K4, K10	X257	no	yes	no
Mains2(DMS) 230VAC L2	F1, F9	K9, K10	X257	yes	yes	yes
DCC FAN 230VAC L3	F1, F10	K5	X259	no	yes	no
DCC D451 24VDC	F12	K5	X259	no	yes	no
WCS 230VAC	F1, F14	K6	X260	no	yes	no
PHS 1(Vert.FC, Motor) 230VAC L2	F1, F8, F15	K4, K10 (or S1)	X257	no	yes	no
PHS 2(PT_PWR G700) 230VAC L3	F1, F8, F16	K4, K10 (or S1)	X257	no	yes	no

Service switch

The PDS has the following service switches and corresponding LED's (lamps).



Fig. 5: Service switch

Tab. 2 Service switches and LEDs

Switch	Lamp	Function
S1	H1 on, if service function via S1 is activated.	Provides power to the couch when system power is switched off; the DMS is still under power. Enables the couch to be moved by the service switches in the table base. Collision protection is disabled! This allows the couch to be used as a lifting device while replacing components of the rotating gantry.
S2	H2, on, if service function via S2 is activated.	Switches the system to Comp_on; the DMS is still under power. It is not possible to switch on the power supply for the gantry and the generator with Sys_on in the Control box.

WARNING

[hz_serdoc_F13G01U11M01]

Use of service-enable switch (table) disables collision protection.

Risk of accident and injury!

⇒ **Be careful when moving the table.**

Other components

The PDS has the following other components.

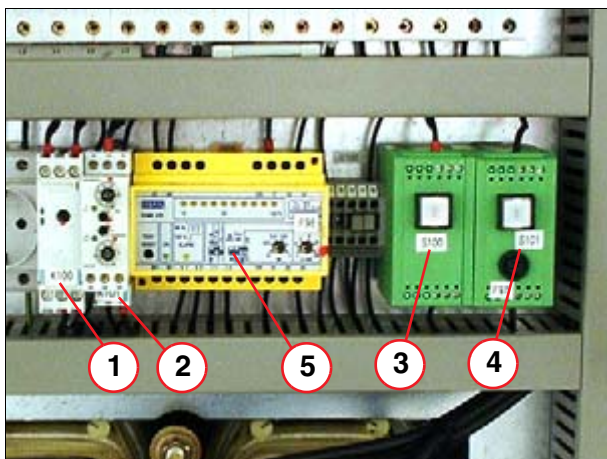


Fig. 6: Other components

Pos. 1	K100
Pos. 2	K101
Pos. 3	S100
Pos. 4	S101
Pos. 5	F98

F98 Leakage current monitor

F98 is a leakage current monitor device (5/Fig. 6 / p. 14). It monitors the 3 input phases of the system via T101 (current transformer) for symmetry. The resulting current of the 3 phases (to and from the system) must be zero. If the resulting current exceeds a certain value the power supply is switched off via the current circuit breaker F99.

Tab. 3 LED's F98 Leakage current monitor

LED	Normal status	Function
ON	ON	ON-LED on shows that the monitor is working.
ALARM	OFF	Alarm LED brinks if 50% of the leakage current limit is reached.
LED bar	OFF	LED bar indicates the leakage current level related to the set response value.

K100 / K101

K100 / K101(1 and 2/ Fig. 6 / p. 14)

Tab. 4 LED's K100/K101

LED	green (230 V)	yellow	yellow
K100 Time delay relay	n. a.	OFF	OFF
K101 Current control relay (DCC T1)	ON	ON	n. a.

S100 / S101

The service switch S100 and S101 (3 and 4/ Fig. 6 / p. 14) can be used to

- S100 = Function test for leakage current measured via current loop T1 (optional).
- S101 = Function test for leakage current measured via current loop T101.

Please refer (Main breaker functionality test / p. 15)

Main breaker functionality test

Fig. 7: F99

NOTE

If fuse F9 is switched off, the main breaker F99 will not work.

1. Switch on the power using the onsite ON/OFF switch.
 - ⇒ Make sure all computer (ICS, IRS and IES) is switched off.
 - ⇒ Make sure F9 is switched on otherwise the main breaker F99 will not work.
2. Check the status of the K100 and K101 LED"s (Tab. 4 / p. 14).

If one of the LED"s has the wrong status, switch the power off and on using the onsite ON/OFF switch. If the problem persists, check all the wiring.
3. If the LED"s status is o.k., press the S100 test key (3/ Fig. 8 / p. 16) and hold for at least 2 seconds.
 - ⇒ The main breaker F99 must trip off to the middle position.
4. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).
5. Check the status of the K100 and K101 LED"s (Tab. 4 / p. 14).

If one of the LED"s has the wrong status, switch the power off and on using the onsite ON/OFF switch. If the problem persists, check all the wiring.
6. **Leakage Current Test** // If the LED"s status is o.k., press the S101 test key (4/ Fig. 8 / p. 16) and hold for at least 2 seconds.
 - ⇒ The main breaker F99 must trip off to the middle position.

7. To switch on the main breaker F99, first move the breaker to the OFF position and then to the ON position (requires extended pressure).

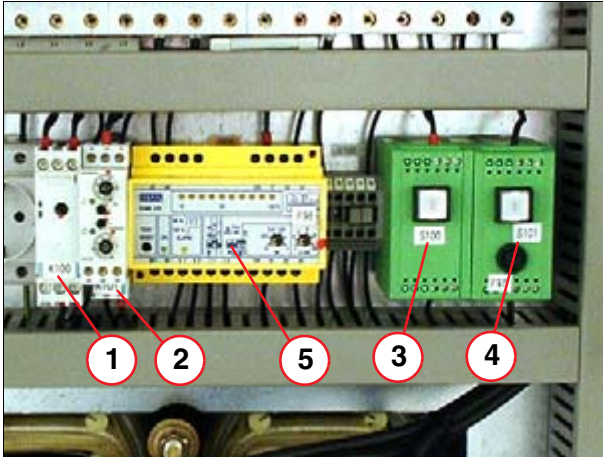


Fig. 8: Other components

Pos. 1	K100
Pos. 2	K101
Pos. 3	S100
Pos. 4	S101
Pos. 5	F98

Additional hints

- **Fuse:**

If one of the F100 fuses is defective, all three fuses have to be replaced!

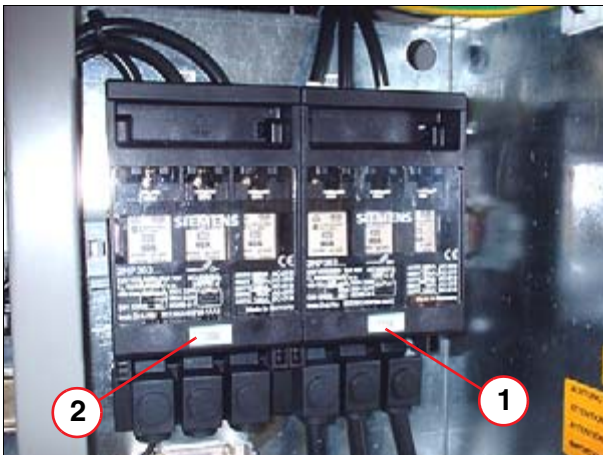


Fig. 9: F100 and F200

Pos. 1	F100
Pos. 2	F200

- **UPS:**

For troubleshooting purposes, you may bypass the UPS. Therefore, you can shortcut the in- and outgoing power plugs of the UPS.

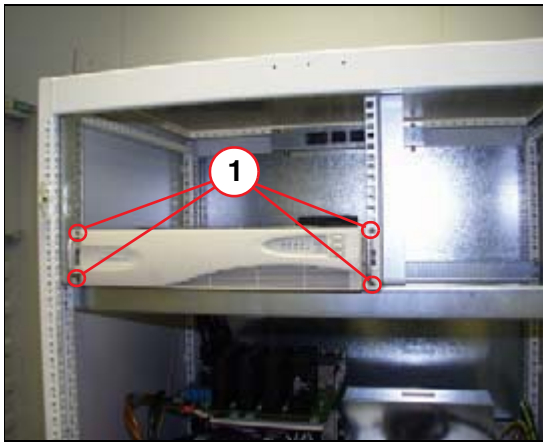


Fig. 10: UPS- uninterruptible power supply

Pos. 1 4 x fastening screws

Chapter	Change	Reason
FMP	SOMATOM 40 was added	136660 (P30G)

H

hz_serdoc_F13G01U03M01	4
hz_serdoc_F13G01U05M03	5
hz_serdoc_F13G01U11M01	5, 13
hz_serdoc_F13G01U12M03	4

